

A Geometric Rotation Heuristic for Quadratic Inequalities

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Abstract

This paper presents a geometric heuristic for determining the solution set structure of quadratic inequalities. By rotating the inequality symbol 90° counter-clockwise, its orientation can be compared directly to the concavity of the parabola. Alignment between the rotated symbol and the curve indicates a single bounded interval, whereas a mismatch indicates two disjoint intervals. We formalise this spatial shortcut using second-derivative concavity analysis.

1 Introduction

When solving quadratic inequalities of the form $ax^2 + bx + c < 0$ or $ax^2 + bx + c > 0$, students must determine whether the critical values bound a single interval (e.g., $x_1 < x < x_2$) or two disjoint intervals (e.g., $x < x_1$ or $x > x_2$). While standard algebraic methods and sketches are effective, spatial recognition shortcuts can accelerate verification.

2 Formalisation and 4-Case Analysis

Let $f(x) = ax^2 + bx + c$ be a quadratic function with real coefficients where $a \neq 0$, having two distinct real roots x_1 and x_2 such that $x_1 < x_2$.

We define the spatial orientation of the parabola $y = f(x)$ as:

- **Convex ($\cup$ -shape):** $a > 0$, opening upwards.
- **Concave ($\cap$ -shape):** $a < 0$, opening downwards.

Applying a 90° counter-clockwise (CCW) rotation to the inequality symbols yields their corresponding geometric visual representations:

$$\begin{aligned} '<' &\xrightarrow{90^\circ \text{ CCW}} 'v' \text{ (opens upwards)} \\ '>' &\xrightarrow{90^\circ \text{ CCW}} '\wedge' \text{ (opens downwards)} \end{aligned}$$

2.1 Case-by-Case Verification

Case 1: $a > 0$ and $f(x) < 0$ The graph is $\cup$ -shaped. The inequality sign ' $<$ ' rotates to ' v '. Both the curve and the rotated symbol open upwards (**Match**). The function values lie strictly below the x -axis between the roots, yielding a single bounded interval:

$$x_1 < x < x_2 \tag{1}$$

Case 2: $a > 0$ and $f(x) > 0$ The graph is $\cup$ -shaped. The inequality sign ' $>$ ' rotates to ' $\wedge$ '. The curve opens upwards while the symbol opens downwards (**Mismatch**). The function values lie above the x -axis outside the roots, yielding two disjoint intervals:

$$x < x_1 \quad \text{or} \quad x > x_2 \tag{2}$$

Case 3: $a < 0$ and $f(x) > 0$ The graph is $\cap$ -shaped. The inequality sign ' $>$ ' rotates to ' $\wedge$ '. Both the curve and the rotated symbol open downwards (**Match**). The function values lie above the x -axis between the roots, yielding a single bounded interval:

$$x_1 < x < x_2 \tag{3}$$

Case 4: $a < 0$ and $f(x) < 0$ The graph is $\cap$ -shaped. The inequality sign ' $<$ ' rotates to ' v '. The curve opens downwards while the symbol opens upwards (**Mismatch**). The function values lie below the x -axis outside the roots, yielding two disjoint intervals:

$$x < x_1 \quad \text{or} \quad x > x_2 \tag{4}$$

3 Calculus Basis for Concavity

The spatial concavity of $f(x) = ax^2 + bx + c$ can be rigorously derived using its second derivative:

$$\frac{d^2}{dx^2} [ax^2 + bx + c] = 2a$$

Since $2a$ is constant, the sign of a determines global concavity:

- $f''(x) = 2a > 0 \implies$ Strictly convex ($\cup$ -shape, local minimum).
- $f''(x) = 2a < 0 \implies$ Strictly concave ($\cap$ -shape, local maximum).

4 Worked Numerical Example

Consider solving the quadratic inequality:

$$x^2 - 3x - 10 < 0$$

1. **Identify Orientation:** Here $a = 1 > 0 \implies f''(x) = 2 > 0$ (convex, $\cup$ -shape).
2. **Find Roots:** Factoring yields $(x - 5)(x + 2) = 0$, giving critical values $x_1 = -2$ and $x_2 = 5$.
3. **Apply Heuristic:** The inequality sign ' $<$ ' rotates 90° CCW to ' $\vee$ ' (opens upwards).
4. **Evaluate Alignment:** Curve ($\cup$) and symbol (' $\vee$ ') match. A match dictates a single bounded interval: $-2 < x < 5$.

5 Summary of Cases

Parabola	Inequality	Rotated Symbol	Alignment	Solution Structure
$a > 0$ ($\cup$)	$<$	$\vee$ (up)	Match	Bounded: $x_1 < x < x_2$
$a > 0$ ($\cup$)	$>$	$\wedge$ (down)	Mismatch	Disjoint: $x < x_1$ or $x > x_2$
$a < 0$ ($\cap$)	$>$	$\wedge$ (down)	Match	Bounded: $x_1 < x < x_2$
$a < 0$ ($\cap$)	$<$	$\vee$ (up)	Mismatch	Disjoint: $x < x_1$ or $x > x_2$

Table 1: Classification table for the 90° CCW rotation heuristic.

6 Conclusion

This spatial heuristic offers a rapid visual check for secondary school students. Grounding the heuristic in second-derivative calculus ensures the shortcut maintains mathematical rigor without overcomplicating the underlying algebra.